

Probability and Statistics

Lecture 6 - Bernoulli's Trials and Binomial Distribution

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Bernoulli's Trials

We will learn:

- ▶ what is Bernoulli's Trials
- ▶ Its importance to determine probability distribution of random variable

Bernoulli's Trials - Simple Definition

Any trial of random experiment is called Bernoulli's trial if it satisfies following conditions:

1. Trial should be finite
2. Trials should be independent (probability of one trial should not depend on other trial)
3. There is only two outcomes: success and failure
4. The probability of success in each trial is same

Note: if p = Probability of success in a trial, then:

1- $p = q$, where q = probability of failure in a trial

Bernoulli's Trials - Example 1

You toss a coin 10 times (each time you toss the coin is a trial).

1. number of trials is 10 (i.e. finite).
2. Every trial is not depending on other trials (i.e. they are independent trials).
3. Outcome of each trial could only be Head or Tail. Suppose success is to get a Head. If you don't get Head then you will only get Tail that represents failure (only two possible outcomes: success or failure).
4. In every trial the probability to get success is $\frac{1}{2}$ (means probability of success in each trial is same)

Hence, every trial of this experiment is a Bernoulli's Trials as it satisfies the 4 conditions of Bernoulli's trial.

Bernoulli's Trials - Example 2

There is a bag that contains 5 white balls and 3 black balls. You have to take out 1 ball from the bag. Success is to get a white ball. You are asked to perform 6 trials. Is this a valid example of Bernoulli's trials in case of (1) replacement and (2) without replacement.

(1) $p = \text{probability of success} = \frac{5}{8}$, $q = \text{probability of failure} = \frac{3}{8}$

- ▶ Due to replacement, probability of success will always be same.
- ▶ Every trial is independent.
- ▶ Ball is either white or black (only two outcome: success or failure).
- ▶ Total number of trials is 6 i.e. finite.

So, it is a valid example of Bernoulli's trial.

Bernoulli's Trials - Example 2

There is a bag that contains 5 white balls and 3 black balls. You have to take out 1 ball from the bag. Success is to get a white ball. You are asked to perform 6 trials. Is this a valid example of Bernoulli's trials in case of (1) replacement and (2) without replacement.

(2) In first trial, $p = \text{probability of success} = \frac{5}{8}$, $q = \text{probability of failure} = \frac{3}{8}$

in second trial, $p = \frac{4}{7}$ (if you picked a white ball in 1st trial)

Due to non-replacement trial, probability of success will not be same in all trials.

So, it is not a valid example of Bernoulli's trial.

Probability Distribution for Bernoulli's Trials

Suppose you are performing a random experiment that satisfies all the conditions of Bernoulli's trials.

Let p = probability of success in trial

q = probability of failure in trial

Let experiment repeat n times, then the probability of r -successes in n trials: $P(X=r) = {}^nC_r \cdot p^r \cdot q^{n-r}$

If you have Random variable X and you want to find probability distribution of X , use the above formula by using values in the domain of X . For example,

$$X = 0, 1, 2, \dots$$

$$P(X) = {}^nC_0 \cdot p^0 \cdot q^{n-0}, {}^nC_1 \cdot p^1 \cdot q^{n-1}, {}^nC_2 \cdot p^2 \cdot q^{n-2}, \dots$$

And it is called Binomial Distribution.

Binomial Probability Distribution is a type of Discrete Probability Distribution in which value of r must be Discrete.

Probability Distribution for Bernoulli's Trials

Example:

Suppose you are performing a random experiment of tossing a coin and you want to repeat it 10 times. What will be the probability to get Head in 4 trials out of 10 trials..

$n=10$ (because we have 10 trials)

$p = \frac{1}{2}$ = probability of success in a trial (success is to get a Head)

$q = \frac{1}{2}$ = probability of failure in trial

$$P(X=r) = {}^nC_r \cdot p^r \cdot q^{n-r}$$

$$P(X=4) = {}^{10}C_4 \cdot p^4 \cdot q^{10-4}$$

Probability Distribution for Bernoulli's Trials

Example:

$$\begin{aligned}P(X=4) &= {}^{10}C_4 \cdot p^4 \cdot q^{10-4} \\&= {}^{10}C_4 \cdot \left(\frac{1}{2}\right)^4 \cdot \left(\frac{1}{2}\right)^6 \\&= \frac{10!}{4!(10-4)!} \cdot \left(\frac{1}{2}\right)^4 \cdot \left(\frac{1}{2}\right)^6 \\&= \frac{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6!}{4!(6)!} \cdot \left(\frac{1}{2}\right)^4 \cdot \left(\frac{1}{2}\right)^6 \\&= \frac{10 \cdot 9 \cdot 8 \cdot 7}{4 \cdot 3 \cdot 2 \cdot 1} \cdot \left(\frac{1}{2}\right)^4 \cdot \left(\frac{1}{2}\right)^6 \\&= \frac{10 \cdot 9 \cdot 8 \cdot 7}{4 \cdot 3 \cdot 2 \cdot 1} \cdot \left(\frac{1}{2}\right)^{10}\end{aligned}$$

Exercise-1

A die is thrown 6 times. If getting an odd number is a success, what is the probability of (1) 5 success, (2) at least 5 success, and (3) at most 5 success?

Let $n = 6$

$$p = \frac{3}{6} = \frac{1}{2} \quad p = \text{probability of success}$$

$$q = \frac{3}{6} = \frac{1}{2} \quad q = \text{probability of failure}$$

$$r = 5$$

Now, by Binomial distribution, the probability of r -success:

$$P(X=r) = {}^nC_r \cdot p^r \cdot q^{n-r} = {}^nC_r \cdot \left(\frac{1}{2}\right)^r \cdot \left(\frac{1}{2}\right)^{n-r}$$

$$P(X=5) = {}^6C_5 \cdot p^5 \cdot q^{6-5}$$

Exercise-1

$$P(X=r) = {}^nC_r \cdot p^r \cdot q^{n-r} = {}^6C_r \cdot \left(\frac{1}{2}\right)^r \cdot \left(\frac{1}{2}\right)^{6-r}$$

(1) $r=5$ (5-success), Find $P(5\text{-success})$

$$\begin{aligned} P(X=5) &= {}^6C_5 \cdot \left(\frac{1}{2}\right)^5 \cdot \left(\frac{1}{2}\right)^{6-5} \\ &= \frac{6!}{5!(6-5)!} \cdot \left(\frac{1}{2}\right)^5 \cdot \left(\frac{1}{2}\right)^1 \\ &= \frac{6!}{5!(1)!} \cdot \left(\frac{1}{2}\right)^6 \\ &= \frac{6 \cdot 5!}{5! \cdot 1} \cdot \left(\frac{1}{2}\right)^6 \\ &= 6 \cdot \left(\frac{1}{2}\right)^6 \\ &= 6 \cdot \frac{1}{64} \\ &= \frac{3}{32} \end{aligned}$$

Exercise-1

$$P(X=r) = {}^nC_r \cdot p^r \cdot q^{n-r} = {}^6C_r \cdot \left(\frac{1}{2}\right)^r \cdot \left(\frac{1}{2}\right)^{6-r}$$

(2) Where $r \geq 5$ (at least 5-success)

$$P(\text{at least 5-success}) = P(X \geq 5) = P(X=5) + P(X=6)$$

$$\begin{aligned} &= {}^6C_5 \cdot \left(\frac{1}{2}\right)^5 \cdot \left(\frac{1}{2}\right)^{6-5} + {}^6C_6 \cdot \left(\frac{1}{2}\right)^6 \cdot \left(\frac{1}{2}\right)^{6-6} \\ &= \frac{6!}{5!(6-5)!} \cdot \left(\frac{1}{2}\right)^5 \cdot \left(\frac{1}{2}\right)^1 + \frac{6!}{6!(6-6)!} \cdot \left(\frac{1}{2}\right)^6 \cdot \left(\frac{1}{2}\right)^0 \\ &= \frac{6!}{5!(1)!} \cdot \left(\frac{1}{2}\right)^6 + \frac{6!}{6!(0)!} \cdot \left(\frac{1}{2}\right)^6 \\ &= \frac{6 \cdot 5!}{5! \cdot 1} \cdot \left(\frac{1}{2}\right)^6 + 1 \cdot \left(\frac{1}{2}\right)^6 \\ &= 6 \cdot \left(\frac{1}{2}\right)^6 + \left(\frac{1}{2}\right)^6 \\ &= 6 \cdot \frac{1}{64} + \frac{1}{64} \\ &= \frac{3}{32} + \frac{1}{64} \\ &= \frac{6+1}{64} = \frac{7}{64} \end{aligned}$$

Exercise-1

$$P(X=r) = {}^nC_r \cdot p^r \cdot q^{n-r} = {}^6C_r \cdot \left(\frac{1}{2}\right)^r \cdot \left(\frac{1}{2}\right)^{6-r}$$

(3) Where $r \leq 5$ (at most 5-success)

$$P(\text{at most 5-success}) = P(X \leq 5) = P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) + P(X=5)$$

Since sum of all probabilities = 1, i.e. $P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) + P(X=5) + P(X=6) = 1$

$$P(X=0) + P(X=1) + P(X=2) + P(X=3) + P(X=4) + P(X=5) = 1 - P(X=6)$$

$$\text{So, } P(X \leq 5) = 1 - P(X=6)$$

$$= 1 - {}^6C_6 \cdot \left(\frac{1}{2}\right)^6 \cdot \left(\frac{1}{2}\right)^{6-6}$$

$$= 1 - \frac{6!}{6!(6-6)!} \cdot \left(\frac{1}{2}\right)^6 \cdot \left(\frac{1}{2}\right)^0$$

$$= 1 - \frac{6!}{6!(0)!} \cdot \left(\frac{1}{2}\right)^6$$

$$= 1 - 1 \cdot \left(\frac{1}{2}\right)^6$$

$$= 1 - \left(\frac{1}{2}\right)^6$$

$$= 1 - \frac{1}{64}$$

$$= \frac{64-1}{64} = \frac{63}{64}$$

Exercise-2

A pair of dice is thrown 4 times. If getting same number on both dice is success. Find the probability of 2-successes.

$$n = 4, r = 2$$

Success: both dice show same number so for one trial:

$$p = \frac{6}{36} = \frac{1}{6}$$

$$q = 1 - p = 1 - \frac{1}{6} = \frac{5}{6}$$

By Binomial Distribution, probability of $X=r$ success is:

$$P(X=r) = {}^nC_r \cdot p^r \cdot q^{n-r}$$

$$\begin{aligned} P(X=2) &= {}^4C_2 \cdot p^2 \cdot q^{4-2} = \frac{4!}{2!(4-2)!} \cdot \left(\frac{1}{6}\right)^2 \cdot \left(\frac{5}{6}\right)^2 = \frac{4 \cdot 3 \cdot 2!}{2!(2)!} \cdot \left(\frac{1}{6}\right)^2 \cdot \left(\frac{5}{6}\right)^2 = \frac{4 \cdot 3}{2 \cdot 1} \cdot \left(\frac{1}{36}\right) \cdot \left(\frac{25}{36}\right) \\ &= 6 \cdot \frac{1}{36} \cdot \frac{25}{36} = \frac{25}{216} \end{aligned}$$